Solution

Let AB be the height of the TV tower.

$CD = 20$ m.

Let BC be the width of the canal.

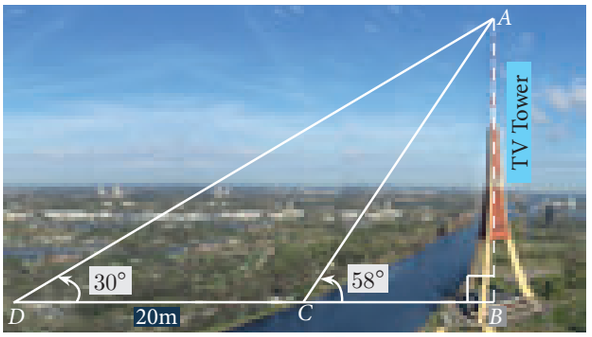


Fig. 6.17

In the right angled ΔABC, $\tan 58° = \dfrac{AB}{BC}$

$$1.6003 = \frac{AB}{BC} \qquad \ldots(1)$$

In the right angled ΔABD, $\tan 30° = \dfrac{AB}{BD} = \dfrac{AB}{BC + CD}$

$$\frac{1}{\sqrt{3}} = \frac{AB}{BC + 20} \qquad \ldots(2)$$

Dividing (1) by (2) we get, $\dfrac{1.6003}{\frac{1}{\sqrt{3}}} = \dfrac{BC + 20}{BC}$

$$BC = \frac{20}{1.7717} = 11.29\,\text{m} \qquad \ldots(3)$$

$$1.6003 = \frac{AB}{11.29} \text{ [from (1) and (3)]}$$

$$AB = 18.07$$

Hence, the height of the tower is 18.07 m and the width of the canal is 11.29 m.

Example 6.24 An aeroplane sets off from G on a bearing of $24°$ towards H, a point 250 km away. At H it changes course and heads towards J deviates further by $55°$ and a distance of 180 km away.

(i) How far is H to the North of G?

(ii) How far is H to the East of G?

(iii) How far is J to the North of H?

(iv) How far is J to the East of H?

$$\begin{pmatrix} \sin 24° = 0.4067 & \sin 11° = 0.1908 \\ \cos 24° = 0.9135 & \cos 11° = 0.9816 \end{pmatrix}$$

Solution

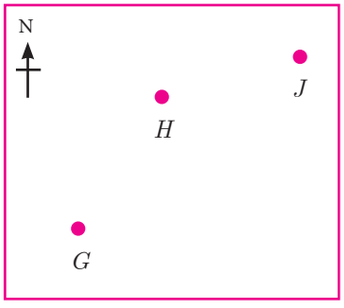


Fig. 6.18 (a)

(i) In the right angled ΔGOH, $\cos 24° = \dfrac{OG}{GH}$

$$0.9135 = \frac{OG}{250};\quad OG = 228.38 \text{ km}$$

Distance of H to the North of $G = 228.38$ km

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